Who Gets the Recovery Hour?

Data Centre Flexibility and Small-State Agency across Indo-Pacific Power Borders

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Abstract

Can a government gain control over digital infrastructure by making its electricity demand interruptible? This paper argues that the answer depends on who governs the recovery of deferred computation. Data centre efficiency standards, demand-response programmes and cross-border electricity agreements regulate different parts of the same service obligation. A structured comparison of five official documents from Singapore, Malaysia and the Singapore-Australia electricity partnership identifies these distinct control points without treating policy announcements as demonstrated resilience outcomes. We develop a bilateral recovery test: admit new compute only when its fixed demand and deadline-bound deferred work can be served while protecting specified essential electricity loads in both exporting and importing systems. A reproducible two-system model translates that institutional commitment into an executable admission constraint. In a constructed 24-hour stress system, the test admits 5.24 MW of average IT demand, compared with 2.15 MW under inflexible operation. However, a daily energy bound suggests 7.04 MW, and omitting the exporter's essential-service floor permits 7.44 MW. These differences arise from recovery timing and the geographical allocation of protection, rather than greater physical supply. Across 288 sensitivity configurations, an independent online scheduling oracle agrees with the optimisation. The contribution is a governance diagnostic, not a new scheduling algorithm or a regional capacity forecast. Small and middle powers can exercise agency by negotiating jointly protected service floors, auditable workload deadlines and recovery obligations at the interface between digital investment and electricity trade. Such arrangements make integration more governable; efficiency labels and nominal interruptibility alone cannot establish the same commitment.

1. The politics of the hour after curtailment

An electricity shortage can be relieved by postponing computation. But deferred work usually returns. The political bargain therefore extends beyond who can order a data centre to reduce its demand: it also determines whose electricity service takes priority when that centre catches up. For a state importing power, an apparently flexible digital investment can remain dependent on another jurisdiction's willingness and ability to supply the recovery period. For an exporter, supporting its neighbour's digital economy can create a competing claim on scarce electricity. Neither outcome follows automatically from interconnection. Both depend on how protection and completion obligations are arranged.

This question joins two domains often governed separately: data centres and cross-border power grids. Singapore's 2024 Green Data Centre Roadmap linked additional capacity to efficiency and green energy [8]. Malaysia's sustainable-development guideline made resource metrics relevant to a specified investment-incentive pathway [10]. Singapore's demand-side reforms addressed dispatchable demand, while the Singapore-Australia electricity framework addressed regional coordination [9,12]. These initiatives demonstrate agency through rule design and conditional market access. Their objects of control nevertheless differ: efficient operation, investment eligibility, short-run response and international cooperation are distinct commitments.

The argument is that small-state agency includes the ability to define an enforceable joint recovery obligation. A government need not own the cloud, manufacture accelerators or domestically generate every

electron to influence the terms under which digital demand uses an interconnected electricity system. Conversely, possessing the power to interrupt demand does not establish that the resulting backlog can be completed without shifting scarcity elsewhere. The relevant question is whether integration preserves agreed service protections through the entire workload cycle.

The bilateral recovery test asks how much compute can be admitted while meeting deadlines and both electricity service floors. Institutional observations, formal implications and numerical illustrations are distinguished throughout. The contribution bridges governance and existing scheduling methods: protection becomes an explicit input, and the binding recovery interval becomes a reviewable output that digital and energy authorities can jointly specify and negotiate.

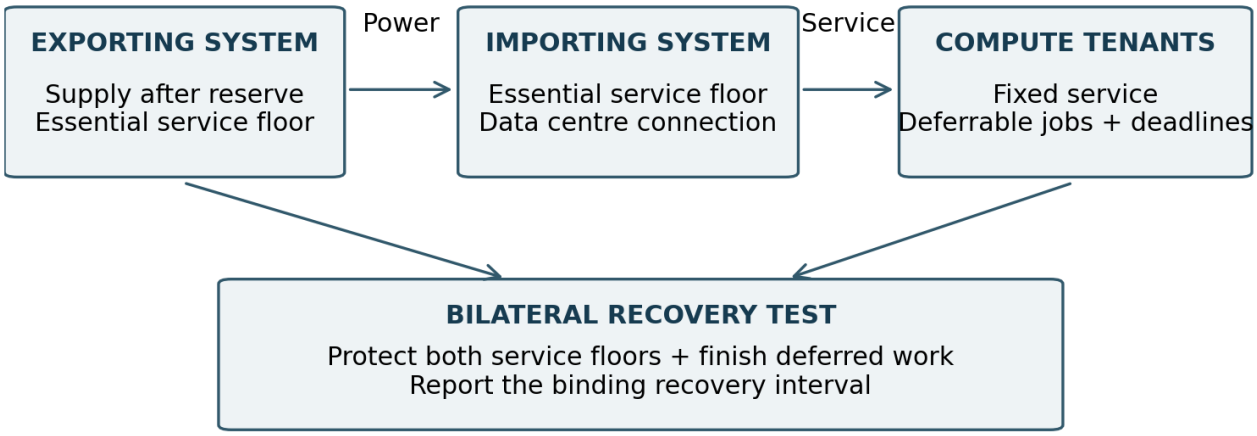


Figure 1. The proposed governance interface. Investment approval, power dispatch and tenant completion obligations must be assessed together. The diagram represents the analytical framework, not a description of an existing regional dispatch authority.

2. What is established, and what this study adds

Three bodies of work frame the problem. Weaponized interdependence explains how authority over central network nodes can create coercive advantages [1]. Regime complexity directs attention to interactions among partially overlapping institutions [2]. Here, the mechanism is a mismatch between the actors approving digital investment, those dispatching electricity and those promising compute completion. Their separate decisions jointly determine a service obligation even when no institution explicitly evaluates that obligation across borders. Dependence concerns ordinary allocation during scarcity as well as deliberate denial; the model does not presume geopolitical coercion.

Technical research has long considered data centre demand response. Wang and colleagues modelled interactions between utilities and geographically distributed computation [3]. Norris and colleagues analysed curtailment-enabled headroom for large flexible loads [4]. Williams and colleagues reported power-flexibility experiments on a GPU cluster and geographical workload shifting [6]. These studies preclude claiming the invention of flexible computing, grid-aware scheduling or curtailment-enabled capacity. Our workload abstraction is less detailed than a cluster controller and cannot establish achievable flexibility for a particular application.

The institutional gap is also narrower than a general call for better contracts. Luo, Tian and Guo explicitly connect deployability to economic incentives, leasing arrangements and operational control [5]. The IEA's October 2026 review surveys 34 policy jurisdictions and discusses connection allocation, cost responsibility and credible flexibility commitments, including dispatch, duration and verification [7, pp. 3, 63-65]. The present study builds on those insights by asking which geographical service floors and recovery deadlines a cross-border admission decision protects. It supplies a reproducible answer for a declared model, rather than

suggesting that contractual credibility was previously overlooked.

The contribution is an institutional application with an inspectable scarcity certificate: combine exporter and importer protections with conservation of deferred work, then identify the interval that makes admission fail. Identical physical systems can protect different users or promise different completion dates. Searches through 17 October 2026 covered flexible-load headroom, compute-electricity contracting, energy justice and cross-border digital infrastructure. They identified substantial adjacent work, including [5,7]; the proposed contribution concerns the joint protection and recovery commitment.

3. Comparative policy evidence: five documents, different control points

The comparison covers five official documents: Singapore's roadmap factsheet and October 2025 flexibility announcement; Malaysia's sustainable-development guideline and September 2026 coordination announcement; and the October 2025 Singapore-Australia electricity framework [8-12]. Selection follows a data centre from investment approval to electricity response and regional supply. These are purposively selected policy functions, not a representative country sample.

Each document is coded for institutional status, decision-maker, control object, temporal commitment and inputs needed for bilateral completion assessment. The source ledger retains locators and separates explicit provisions from unreported terms. Missing terms do not establish absent legal rights or policy failure. The comparison identifies governance choices and information requirements, rather than estimating causal effects on outages.

Document / status	Control point	Additional recovery question
Singapore roadmap [8] Planning factsheet	Capacity and resource performance	Which work can be dispatched?
EMA flexibility [9] Reform announcement	Activation and availability	When must deferred work finish?
Malaysia guideline [10] DESAC eligibility	Design resource metrics	Who controls tenant workloads?
MIDA task force [11] Coordination announcement	Investment applications	Which joint test is applied?
Australia-Singapore [12] Non-binding framework	Trade and governance	Whose service has priority?

Table 1. Five documents, different institutional objects. Questions identify additional inputs; they do not demonstrate an absence of rights in other instruments. Locators and coding boundaries are in the public evidence ledger.

Singapore's roadmap illustrates selective expansion through industry partnerships and resource performance. Its stated additional-capacity ambition is a planning commitment, not a measurement of demand that can be called off during a shortage [8, paras. 1-2]. Efficiency remains valuable: a lower facility overhead increases the compute supportable by a fixed electricity envelope. But an efficiency ratio contains neither a dispatch right nor a promise about a workload's completion date. A planning authority therefore needs a second assessment if it wants expansion to provide dependable scarcity response.

EMA's October 2025 announcement offers a more direct temporal distinction. It planned to reduce each Interruptible Load activation to 30 minutes, with implementation details to follow, and described existing participation as opportunistic because production schedules constrain availability [9, paras. 3-6]. The announcement also distinguished price-driven Demand Response from imbalance response. It is evidence that Singapore recognises the difference between technical resources and consistently available

commitments. It is not evidence that a proposed 30-minute activation discharges a data centre's later compute obligations, or that the announced change had already taken effect. Activation duration and workload recovery must be assessed separately.

Malaysia's guideline identifies PUE, CUE and WUE, with category-specific design targets and an explicit application scope: DESAC tax-incentive applications received by 31 December 2027 [10, sections 2-3 and Appendix 1]. The text uses different normative terms for different provisions; it should not be converted into a universal statutory operating standard. Its importance here is the investment lever: eligibility can shape operator behaviour before scarce resources are committed. MIDA's September 2026 announcement adds a coordination lever through a task force co-chaired by MITI and the Ministry of Digital, with MIDA as the focal point for new and expansion applications [11, concluding section]. This can bring digital investment and resource planning into one process. The announcement alone does not establish a common dispatch protocol or verified compliance.

The Singapore-Australia framework provides a regional venue for that coordination. Its principles address oversight and accountability; its governance section recognises stakeholder consultation. It also favours prior engagement over arbitrary trade restrictions while expressly stating that the framework does not create legally binding rights or obligations [12, Ten Principles; sections II-III; Duration and signature]. It supplies an institutional basis for negotiating predictable allocation rules, rather than an already operative guarantee of emergency electricity. This distinction matters: cross-border reliability cannot be inferred from cooperative intent, and emergency allocation should not be improvised as a unilateral exception after investment is sunk.

Across these documents, agency is visible in selection, incentives, coordination and agenda-setting. The evidence does not justify ranking national resilience. Instead, it identifies an interface problem: the reviewed materials do not supply a complete joint specification of essential-service floors, dispatchable workloads and recovery deadlines. That is a tractable object for subsequent bilateral or project-level negotiation, including negotiation by an exporting state that wants local electricity users to benefit from regional digital growth.

4. An executable bilateral recovery test

Consider an exporting electricity system E and an importing system I over hourly periods t. Let S denote available supply after a separately specified operating-reserve allowance, H protected non-data-centre demand, K the transfer limit and eta the delivered fraction of exported power. We assume the baseline is feasible, exports are unidirectional and there is no storage. With both service floors protected, the additional facility power available at I is

$$h_t = S_t^I - H_t^I + \eta \min\{K_t,\ S_t^E - H_t^E\}. \qquad (1)$$

Here h is headroom, not generation. The reduction follows directly: E can export at most its supply remaining after H_E; I must first serve H_I. The model permits spillage and treats each national system as one transport node. It excludes internal congestion, frequency stability, generator ramps and endogenous prices. Its simplicity makes the protection boundary visible but prevents its use as a grid-connection study. Omitting H_E in equation (1) defines an importer-only relaxation. That comparator explicitly permits allocating electricity away from E's protected users; it is not a claim about existing regional contracts.

Let q be average admitted IT demand, p a constant illustrative PUE, and f the flexible fraction. Fixed IT demand is (1-f)q each hour. Four flexible jobs arrive at hours 0, 6, 12 and 18. Each requires 6fq IT MWh

and may execute from release r until d=min(r+6+delta,24), where delta is additional deadline slack. Jobs are divisible and preemptible. Aggregate flexible processing cannot exceed Bfq per hour, with burst factor B=2. Unlike a queue of interactive inference requests, these are declared delay-tolerant workload cohorts. Completion by hour 24 is mandatory; no backlog is hidden beyond the horizon.

Write y_j,t for flexible IT power assigned to job j. The admission programme maximises q subject to nonnegative processing, job windows and the following requirements in every stress scenario:

$$p\left[(1-f)q+\sum_j y_{j,t}\right]\le h_t,\quad \sum_j y_{j,t}\le Bfq,\quad \sum_{t=r_j}^{d_j-1} y_{j,t}=6fq. \qquad (2)$$

The first requirement protects the two electricity floors through equation (1); the last conserves deferred work. Scenarios have no assigned probabilities. Variables for histories with identical observed headroom are constrained to agree up to the current hour, so the optimiser cannot identify a future shock before it becomes observable. These are decision constraints, not forecasts of regional interruption frequency. The test's answer is conditional on the supplied service floors and uncertainty set; it cannot decide whose needs ought to receive protection.

An interval certificate explains rejection without treating the solver as an authority over policy. After serving fixed demand, the hourly flexible capacity is C_t(q)=min(Bfq, h_t/p-(1-f)q), provided the second term is nonnegative. For every interval [a,b), the energy of jobs whose entire executable windows fall inside it must not exceed the sum of C_t(q) over that interval. This is the standard preemptive scheduling feasibility condition, specialised to the protection-adjusted envelope. The admission constraint is an application of established scheduling structure [13].

Necessity follows because those jobs cannot execute outside the interval. For sufficiency, consider the first deadline missed by earliest-deadline-first scheduling and the preceding maximal busy interval for jobs due by that deadline. Its uncompleted work yields an interval violation. Exchanging later-deadline execution with an already released earlier-deadline job preserves feasibility. Thus the online rule requires current capacity, not future supply, in this divisible single-envelope setting. Storage, migration and job-specific limits can invalidate this conclusion.

The code implements a linear programme with shared-history constraints and an independent earliest-deadline-first oracle with capacity bisection. A four-hour witness isolates timing: headrooms (1,1,5,5) and (3,3,3,3) have identical daily energy, yet sequential two-hour job windows, full flexibility and p=1 admit average IT loads of 1 and 3. Completion constraints explain the difference.

5. Results and what they establish

All electrical profiles in the experiment are constructed. E has 28 MW of available supply and an 18 MW protected load. I has 18 MW supply and 16 MW protected demand, rising to 17 MW during hours 18-21. Transfer capacity is 10 MW with eta=0.95. Five scenarios cover normal operation, an early exporter shock, a late exporter shock, an early importer shock and a late joint shock. Exporter shocks remove 6 MW and importer shocks remove 3 MW for six hours, starting at hour 6 or 12. The main configuration uses p=1.3, f=0.8 and delta=6 hours. These values are transparent stress inputs, not estimates fitted to a national grid or an operator's flexibility claims.

Assessment	Average IT MW	Commitment retained / relaxed
Inflexible bilateral	2.154	Both service floors; fixed load
Recovery bilateral	5.244	Both floors; all job windows
Daily energy upper bound	7.045	Total work; timing relaxed
Recovery importer only	7.436	Job windows; exporter floor omitted

Table 2. Main constructed-system results. The last two rows are relaxations of the bilateral recovery commitment, not equivalent service promises. IT MW excludes facility overhead; PUE = 1.3.

Inflexible operation admits 2.15 MW of average IT load. Completing all work within the permitted windows raises admission to 5.24 MW while preserving both electricity floors. The benefit is substantial, but smaller than two relevant relaxations. Counting only total daily headroom yields an upper bound of 7.04 MW, 34.35% above the completion-feasible value. Preserving completion while omitting E's service floor admits 7.44 MW, 41.81% above it. Neither relaxed value is a feasible admission under the bilateral recovery commitment. They identify which promise must change before the larger number can be defended.

The binding interval is particularly informative. In the joint-shock scenario, hours 12-23 provide 81.8 MWh of additional facility electricity. The jobs released at hours 12 and 18 must both finish by hour 24; their associated fixed and flexible demand requires 12pq MWh. Thus q cannot exceed 81.8/(12 times 1.3)=5.24359 MW. The executable schedule attains that bound. Raising capacity above it must add power in that interval, permit later completion, reduce work, or revise protection. More energy earlier in the day cannot complete jobs that have not yet arrived.

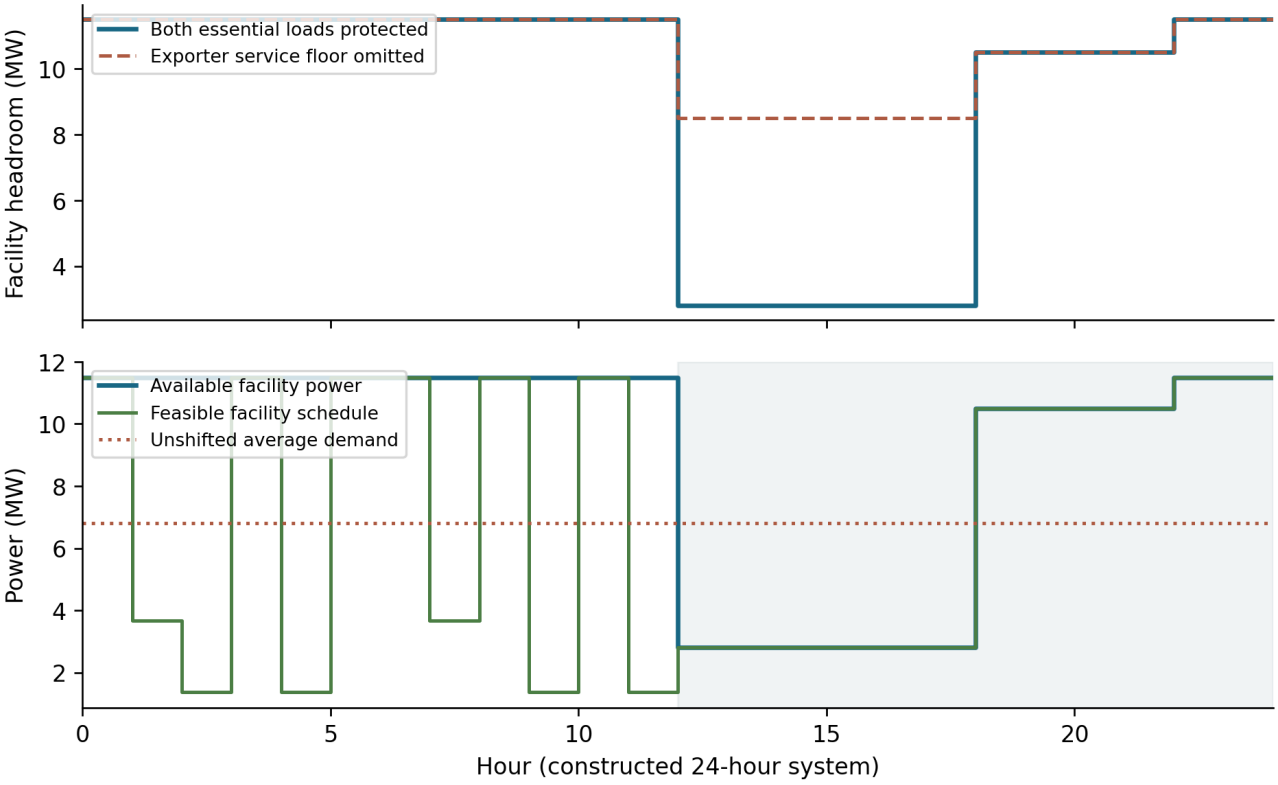


Figure 2. Joint-shock scenario in the constructed system. Top: protection choices change available headroom without changing physical supply. Bottom: one completion-feasible LP schedule at q = 5.24359 MW. The shaded interval [12,24) binds admission. Lines show hourly values; all electrical inputs are illustrative.

A deliberately more permissive cancellation control drops all flexible-work completion requirements. It admits 10.77 MW, but permits cancellation of the flexible fraction rather than postponement. This is not an alternative estimate of equivalent compute service, nor a representation of the cited headroom studies. Its role is to prevent the accounting error of describing permanent demand removal as reversible flexibility. The daily-energy bound in Table 2 retains total-work conservation and is the more informative comparison when recovery timing is the disputed assumption.

The sensitivity grid varies exporter shock severity over 2, 4 and 6 MW; duration over 2, 4 and 6 hours; flexible share over 0.2, 0.4, 0.6 and 0.8; additional slack over 0, 3, 6 and 12 hours; and PUE over 1.2 and 1.4. This gives 288 configurations and 864 sensitivity optimisations, plus five main optimisations. Maximum constraint residual is below 2 times 10^-13. The independent scheduling oracle differs by less than 4.2 times 10^-8 MW across all configurations. Removing the shared-history constraints changes no capacity result, consistent with the scheduling argument. This numerical agreement validates implementation within the model, not real-world power delivery.

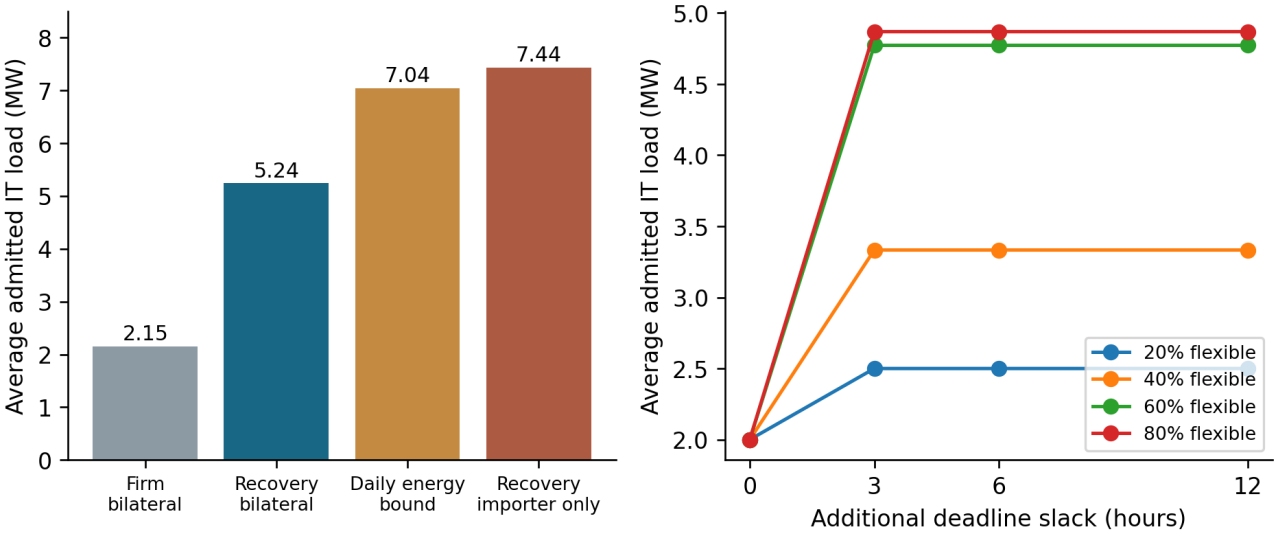


Figure 3. Conditional admission results. Left: main case, PUE 1.3, 80% flexible, six-hour additional slack; the daily-energy and importer-only quantities relax different commitments. Right: sensitivity at six-hour, 6 MW exporter shocks and PUE 1.4; only deadline slack and flexible share vary. These are constructed results, not national capacity estimates.

The sweep also qualifies a simple recommendation to buy longer deadlines. At the severe six-hour shock and PUE 1.4, extending slack from three to six or twelve hours does not increase admission at any tested flexible share. Later cohorts still have to finish within the common study horizon, while fixed demand remains. Deadline flexibility is therefore useful only when it unlocks additional usable capacity. The model gives a definite boundary for the declared commitment; it does not establish an optimal universal deadline.

6. From investment screening to a regional scarcity bargain

The policy implication is to attach a bilateral recovery schedule to relevant digital-capacity and power-trade decisions. This would specify five items: protected service floors in each participating system; workload release and completion commitments; an accountable dispatch chain; available power during recovery; and evidence and compensation rules. Such a schedule is a proposed institutional instrument, not an inferred provision of the reviewed agreements. The model supplies its quantitative core while leaving the choice of beneficiaries and trade-offs to legitimate decision processes.

National authorities can begin at investment screening. The applicant should distinguish continuously required services from work that may be deferred, cancelled or migrated. The energy regulator can test an envelope including recovery. The operator must show that tenant contracts permit the promised response:

an electricity connection agreement cannot itself grant control over customers' computing obligations. Critical digital services belong in the fixed component.

Exporting states can seek protection for local users, transparency about recovery-period exports and agreed exceptions before accepting long-term trade commitments. Importers gain a continuity promise negotiated before scarcity arises. When the interval test fails, parties can evaluate storage, revised deadlines or smaller admission proposals. Coordination can change the allocation bargain; additional electricity requires a physical intervention.

Regional cooperation can standardise the reporting interface without imposing one dispatch authority. A common submission could disclose workload cohorts, hourly residual supply envelopes, reserve assumptions and the smallest failing recovery interval. Governments could publish aggregate certificates while protecting tenant identities and operationally sensitive details. The bilateral framework's existing attention to coordination and consultation provides a plausible venue for proposing such an extension. Its non-binding status also makes clear that operational duties would need separate implementation; the test does not turn a statement of cooperation into an enforceable service right.

Success would require credible information and aligned obligations. Metered activation reductions alone cannot verify that deferred work later finished within the approved envelope. Periodic witnessed tests should reconcile dispatch instructions, facility electricity, completion records and residual backlog. Independent verification could use aggregate cohort accounting rather than customer content. Compensation should distinguish standing availability, actual interruption and agreed recovery, with remedies for misreported flexibility. The paper does not estimate compensation rates or assume that a mathematically feasible promise will be commercially attractive.

Hidden tenant constraints, inflated flexibility, double-counted reserves, congestion or refusal to honour priorities can defeat the arrangement. Monitoring cannot cure an infeasible recovery interval; hardware cannot settle a disputed protection boundary. A commercial decision-support service could help utilities, developers and financiers map obligations, preserve evidence provenance and audit recovery certificates. It would complement existing scheduling tools and power-system studies; commercial demand remains to be tested.

7. Scope and conclusion

The comparison identifies distinct governance levers; the formal analysis translates joint protection and workload conservation into a recovery constraint; the experiments establish its numerical consequences in a declared system. These findings do not identify actual national shortfalls, demonstrate policy failure or establish commercial demand.

Deployment requires project contracts, validated resource envelopes and engineering assessment. The model omits internal networks, storage and dynamics, assumes constant PUE and divisible work, and uses a finite horizon. A rolling implementation needs an explicit terminal backlog condition. These extensions should refine the tested commitment while preserving the visibility of its protection choices.

For the workshop's strategic question, AI-driven digital expansion turns electricity allocation into a condition of infrastructure control. For its agency question, small and middle powers can use investment access, dispatch programmes and bilateral coordination to define who is protected through workload recovery. For its success question, integration becomes more resilient when those obligations are jointly feasible and verifiable; it remains vulnerable when nominal flexibility conceals displaced service or unfinished work. The hour after curtailment is therefore a practical site of regional governance, not merely an operational afterthought.

Data And Code

Code, constructed inputs, policy coding, sensitivity outputs and figure scripts: https://github.com/shi1720/scarcity-reciprocity. All numerical results are reproducible without proprietary data. The repository separates documentary evidence from constructed scenarios.

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